

EDA (Exploratory Data Analysis) Sheet

EDA is the process of understanding, summarizing, and visualizing data before building Machine Learning models.

Complete EDA Checklist

Step	Task	Purpose
1	Load Dataset	Read data into Pandas
2	Check Shape	Rows and columns
3	View Data	Understand records
4	Dataset Information	Data types & null values
5	Statistical Summary	Mean, Median, Std, etc.
6	Check Missing Values	Identify null data
7	Check Duplicates	Find repeated rows
8	Unique Values	Understand categories
9	Univariate Analysis	Analyze single feature
10	Bivariate Analysis	Analyze two features
11	Multivariate Analysis	Analyze multiple features
12	Outlier Detection	Identify extreme values
13	Correlation Analysis	Feature relationships
14	Feature Distribution	Check data spread
15	Categorical Analysis	Analyze categories
16	Feature Engineering Ideas	Create new features
17	Insights & Conclusions	Business understanding

1. Load Dataset

```
import pandas as pd

df = pd.read_csv("data.csv")
```

2. Dataset Overview

```
print(df.head())
print(df.tail())
print(df.shape)
print(df.columns)
```

3. Dataset Information

```
df.info()
```

Output shows:

- Total rows
 - Total columns
 - Data types
 - Missing values
-

4. Statistical Summary

```
df.describe()
```

Shows:

- Mean
 - Median (50%)
 - Standard Deviation
 - Min
 - Max
 - Quartiles
-

5. Missing Values

```
df.isnull().sum()
```

Percentage:

```
(df.isnull().sum()/len(df))*100
```

6. Duplicate Records

```
df.duplicated().sum()
```

7. Unique Values

```
df.nunique()
```

Specific column:

```
df["Gender"].unique()
```

8. Univariate Analysis

Numerical Columns

Histogram

```
import matplotlib.pyplot as plt

df["Age"].hist()
plt.show()
```

KDE Plot

```
import seaborn as sns

sns.kdeplot(df["Age"])
plt.show()
```

Box Plot

```
sns.boxplot(x=df["Age"])
plt.show()
```

Purpose:

- Distribution
 - Skewness
 - Outliers
-

9. Categorical Analysis

Count Plot

```
sns.countplot(x="Gender", data=df)
plt.show()
```

Pie Chart

```
df["Gender"].value_counts().plot.pie(
    autopct='%1.1f%%'
)
plt.show()
```

10. Bivariate Analysis

Numerical vs Numerical

Scatter Plot

```
sns.scatterplot(  
    x="Age",  
    y="Salary",  
    data=df  
)  
plt.show()
```

Finds:

- Positive correlation
 - Negative correlation
 - No correlation
-

Categorical vs Numerical

Bar Plot

```
sns.barplot(  
    x="Department",  
    y="Salary",  
    data=df  
)  
plt.show()
```

Categorical vs Categorical

Crosstab

```
pd.crosstab(  
    df["Gender"],  
    df["Department"]  
)
```

11. Multivariate Analysis

Pair Plot

```
sns.pairplot(df)  
plt.show()
```

Purpose:

- Compare all numerical columns
-

12. Correlation Analysis

Correlation Matrix

```
corr = df.corr(numeric_only=True)
print(corr)
```

Heatmap

```
sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm"
)
plt.show()
```

Interpretation:

Value	Meaning
+1	Perfect Positive
0	No Relation
-1	Perfect Negative

13. Outlier Detection

Box Plot

```
sns.boxplot(x=df["Salary"])
plt.show()
```

IQR Method

```
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR

outliers = df[
```

```
(df["Salary"] < lower) |  
(df["Salary"] > upper)  
]  
  
print(outliers)
```

14. Distribution Analysis

Skewness

```
df.skew(numeric_only=True)
```

Interpretation:

- 0 = Symmetric
 - Positive = Right Skewed
 - Negative = Left Skewed
-

15. Feature Relationships

Covariance

```
df.cov(numeric_only=True)
```

Correlation

```
df.corr(numeric_only=True)
```

16. Business Insights Example

Employee Dataset

Questions:

Highest Paid Department

```
df.groupby("Department")["Salary"].mean()
```

Average Age

```
df["Age"].mean()
```

Gender Distribution

```
df["Gender"].value_counts()
```

Employees Per Department

```
df["Department"].value_counts()
```

Full EDA Template

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("data.csv")

# Basic Information
print(df.head())
print(df.shape)
print(df.info())
print(df.describe())

# Missing Values
print(df.isnull().sum())

# Duplicates
print(df.duplicated().sum())

# Correlation
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
)
plt.show()

# Pairplot
sns.pairplot(df)
plt.show()

# Histogram
df.hist(figsize=(10,8))
plt.show()

# Boxplot
for col in df.select_dtypes(include='number'):
    sns.boxplot(x=df[col])
    plt.show()
```

EDA Interview Questions

1. What is EDA?
2. Why is EDA important?
3. Difference between EDA and Data Cleaning?
4. What is Univariate Analysis?
5. What is Bivariate Analysis?
6. What is Multivariate Analysis?

7. What is Correlation?
8. What is Covariance?
9. What is a Heatmap?
10. What is Pairplot?
11. What is Skewness?
12. What are Outliers?
13. How do you detect Outliers?
14. What is Feature Distribution?
15. Why use Histograms?
16. Why use Boxplots?
17. Difference between Correlation and Causation?
18. What insights can EDA provide?
19. What is Data Visualization?
20. What is the first step in EDA?

EDA Flow for Any Project

Load Data → Understand Data → Clean Data → Univariate Analysis → Bivariate Analysis → Multivariate Analysis → Correlation Analysis → Outlier Detection → Visualization → Business Insights → Feature Engineering → Model Building.